

PXE Boot on UniFi

Troubleshooting and fix documentation for native PXE boot on Juniper's UniFi network.

Problem

Target machines on the engineering subnet (192.168.5.x) could not PXE boot natively — only via a USB flash drive with iPXE bootstrap. This repo documents the investigation, root causes, and the fix applied to pc-deploy (192.168.5.141).

Stack

- **pc-deploy** — Windows 11, 192.168.5.141
- **tftpd64** — TFTP + proxy DHCP server (`C:\tftpd64\`)
- **WinPE** — boot image at `C:\tftpd64\EFI\Boot\bootx64.efi`
- **UniFi router** — EFG Router at 192.168.0.1, provides DHCP options 66/67
- **Engineering switch** — TP-Link TL-SG108-M2 (unmanaged)

Root Causes Found

1. **Boot File missing from tftpd64's [DHCP] config** — proxy DHCP had nothing to advertise
2. **SecurityLevel=1 in tftpd32.ini** — blocked TFTP access to `EFI\Boot\bootx64.efi` (subdirectory)
3. **Wrong INI format** — build script wrote custom keys tftpd64 doesn't recognize

All three were fixed on 2026-06-18. See [PXE-FINDINGS.md](#) for full details.

Fix Summary

Both `C:\tftpd64\tftpd64.ini` and `C:\tftpd64\tftpd32.ini` on pc-deploy were updated: - Added `Boot File=EFI\Boot\bootx64.efi` to [DHCP] - Set `SecurityLevel=0` - Set `PXECompatibility=1` - Set `BaseDirectory=C:\tftpd64` (absolute)

Related Projects

- [pc-imaging-server](#) — WinPE build scripts, tftpd64 setup
 - [computer-inventory](#) — network inventory, UniFi API
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